

Problem Definition: Car Rental System

The car rental industry experiences several operational challenges due to inefficient and unstructured management processes. Many car rental businesses struggle with maintaining accurate records of vehicles, customers, and rental transactions. These records are often handled manually or through disconnected systems, which increases the likelihood of errors and inconsistencies.

One of the key problems is the difficulty in tracking vehicle availability in real time. This can result in double bookings, customer dissatisfaction, and loss of business. Additionally, managing customer information without a centralized system can lead to incomplete or misplaced data, making it hard to maintain reliable customer records.

Another major issue is the lack of proper coordination between different aspects of the business, such as rentals, returns, and payments. Without a well-defined structure, it becomes challenging to ensure that all transactions are accurately recorded and linked. This can lead to financial discrepancies and poor accountability.

Furthermore, the absence of a clear system structure makes it difficult to model and understand the relationships between different components of the business, such as customers, vehicles, and rental agreements. This lack of clarity can negatively impact system development, scalability, and maintainability.

Overall, there is a need to clearly define and understand the problem domain of a car rental business, particularly in terms of managing entities and their relationships, to highlight the inefficiencies and challenges currently faced.